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SUPERSEDING
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MIL-D-26001C
26 Jan 1967

MILITARY SPECIFICATION

DEVELOPERS, PHOTOGRAPHIC, ARMED FORCES

This specification is mandatory for use by all Departments and Agencies of the Department of Defense

1. SCOPE

*1.1 This specification covers the requirements of processing formulas for photographic developers which are supplied in dry powder form, properly stabilized, for use in the preparation of solutions for developing photographic films and papers.

*1.2 Classification

*1.2.1 Types. The armed forces developers shall be of the following types:

Type I	High Contrast, Film
Type II	Medium Contrast, Film
Type IV	Rapid Process, High Contrast, Film

2. APPLICABLE DOCUMENTS

*2.1 The following documents, of the issue in effect on the date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein:

SPECIFICATIONS

Federal

O-C-275	Chemicals, Photographic, Single Substance, General Specification For
PPP-P-16	Packaging of Photographic Dry Chemicals, General Specification for

STANDARDS

Military

MIL-STD-105	Sampling Procedures and Tables for Inspection by Attributes

FSC 6750

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Federal

Fed Std No 170 Film, Photographic, Black and White, Classification and Testing Methods

(Copies of specifications, standards, drawings, and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

*3.1 Chemicals. The chemicals used in the processing formula shall be photographic grade conforming to O-C-275. The chemicals used shall be readily soluble in water, when tested in accordance with 4.3.1.

3.1.1 Mixing of chemicals. The dry chemicals in powder form shall be mixed thoroughly to form a homogenous blend before dispensing into unit container. The chemicals shall be free flowing, readily soluble, free from foreign matter, and shall show no signs of discoloration or dark specks. The temperature and relative humdity of the mixing room shall not exceed 75 degrees Fahrenheit (F) and 25 percent relative humidity.

*3.2 Composition of formula. The three types of developer shall have the respective developing chemicals specified in Table 1. Unless otherwise specified (see 6.2), the allowable tolerance of the total net weight of the individual chemicals shall be ± 3 percent. The supplier shall certify that the composition of the developer furnished performs to the appropriate formulation specified in Table 1.

TABLE 1

COMPOSITIONS OF FORMULAS

TYPE I

Chemicals	One Gallon Size (Grams)	Five Gallon Size (Grams)
p-Methylaminophenol Sulfate	7.6	38.0
Hydroquinone	30.3	151.5
Sodium Sulfite (Anhydrous)	340.7	1,703.5
Sodium Carbonate, Monohydrate	198.7	993.5
Potassium Bromide	18.9	94.5
TOTAL NET CHEMICAL WEIGHT*	596.2	2,981.0

TYPE II

Chemicals	One Gallon Size (Grams)	Five Gallon Size (Grams)
p-Methylaminophenol Sulfate	9.5	47.5
Sodium Sulfite, (Anhydrous)	113.4	567.0
Hydroquinone	9.5	47.5
Sodium Metaborate	37.9	189.5
Potassium Bromide	1.9	9.5
TOTAL CHEMICAL WEIGHT*	172.2	861.0

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TYPE IV

Chemicals	One Gallon Size (Grams)	Five Gallon Size (Grams)
Sodium Sulfite, Anhydrous	91.5	457.5
Sodium Carbonate Monohydrate	165.0	825.0
Potassium Bromide	3.8	19.0
Hydroquinone	40.0	200.0
1-Phenyl 3-Pyrazolidone	1.0	5.0
Benzotriazole	0.4	2.0
TOTAL CHEMICAL WEIGHT*	301.7	1,508.5

*To allow this formula to be prepared in a single powder form it may be necessary to add additional water soluble components, otherwise photographically inert, to prevent interaction or decomposition of the active components while in packaged form.

3.3 Performance. The developer shall perform equal to the control solution when tested as specified in 4.3.3.

3.4 Unit sizes. The chemicals shall be packaged in unit sizes sufficient to prepare working solutions of 1.5, or more gallons, as specified by the procuring activity (see 6.2). For packages other than those specified herein, the weight of the chemical composition of the formula shall be extrapolated accordingly. EXAMPLE: To determine the quantity of sodium sulfite needed for a 20 gallon size of Type IV developer, multiply by 20 the amount needed from one gallon size; i.e., 20 x 91.5 grams = 1830.0 grams.

3.5 Storage stability. The dry powders, in the unit package, shall show no evidence of deterioration after storage for a period of 48 hours at 130° ± 2°F.

3.6 Workmanship. Workmanship shall be in accordance with high grade commercial practice covering the manufacture and packaging of photographic materials.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for Inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Sampling. Sampling shall be performed in accordance with the provisions of MIL-STD-105, except where otherwise indicated. For purposes of sampling, an inspection lot for acceptance inspection shall consist of all material of the same type and item submitted for acceptance and delivery at one time.

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*4.3 Testing of the end item. The end item shall be tested for the applicable characteristics as indicated in Table II for each lot presented for acceptance. The lot size shall be expressed in units of primary containers of developer. The sample unit for the dissolving and performance test shall be one primary container. The inspection levels and acceptable quality levels shall be as indicated in Table II.

*4.3.1 Dissolving test. The dry powder obtained from each sample shall be slowly added to 75 percent of the final volume of distilled water at 90°F ± 5°F while stirring with sufficient agitation to keep the chemicals suspended. Stirring shall continue until chemicals are completely dissolved. Sufficient water shall be added to make final volume of developer and stirring shall continue until a uniform solution is obtained. The mixing process shall not exceed 10 minutes. Ten minutes after mixing the chemical solution shall be colorless to light pink in color. Although a slight flocculence of the cooled liquid is permissible, foreign matter in any form, rust, dirt or undissolved chemicals shall be cause for rejection.

*4.3.2 Storage stability test. Unit containers selected at random from each lot shall be stored at 130° ± 2°F for a period of 48 hours. The samples shall then be removed and allowed to cool to 68°F ± 5°F for 2 hours and then examined. The powder shall be dry, with no discoloration, free-flowing, and free from lumps. No rust shall be observed in the interior of the containers. The contents of the containers shall then be subjected to performance test, specified in 4.3.3 (see Table II).

*4.3.3 Performance test. Prepare two developer solutions in distilled water. The chemicals to be used in the first solution shall be obtained from the chemicals being used in production of the developer. This mixture shall be identified as "sample solution". The chemicals used in the second solution shall be prepared in accordance with the formula for type I, II, or IV, as appropriate in Table I. The latter mixture shall be labeled "control solution". In accordance with the procedures described in FED STD 170, expose identical sensitometric images, containing twenty-one steps, with a range in log exposure of 0.0 to 3.0, onto two strips of photographic film taken from the same box or roll. Process one strip in each of the respective prepared solutions in accordance with the procedure described in FED STD No. 170. Compare the sensitometric characteristics of the film developed in the solution prepared according to the specific formula. The gamma of the curves prepared from film developed in the test solution and the control solution shall not vary by more than ± 0.10, base and fog density shall not vary by more than ± 0.02, and D max shall not vary by more than ± 0.10. Tests shall be performed in duplicate on each sample unit. The average of the two determinations on each sample unit shall be used for plotting the sensitometric curve. Speed method selected for each identical process sensitometric image shall not vary more than one-eight of a stop by comparison.

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TABLE II. Instructions for testing - sample units

Characteristic	Spec. reference		Rqmts. applied to		No. determinations per sample unit	Results reported as		Inspection level 3/	AQL in defects per 100 units
	Requirement	Test method	Indiv. unit	Lot avg.		Pass or fail 1/	Numerically to nearest 2/		
Dissolving test	3.1	4.3.1	X	---	1	X	---	S-1	2.5
Performance test	3.3	4.3.3							
Difference in gamma			X	---	Avg. of 2	---	0.01	S-1	2.5
Difference in base and fog density			X	---	Avg. of 2	---	.01	S-1	2.5
Difference in D max			X	---	Avg. of 2	---	.01	S-1	2.5
Storage stability test	3.5	4.3.2							
Condition of powder			X	---	1	X	---	S-1	2.5
Rust			X	---	1	X	---	S-1	2.5
Performance	3.3	4.3.3							
Difference in gamma			X	---	Avg. of 2	---	.01	S-1	2.5
Difference in base and fog density			X	---	Avg. of 2	---	.01	S-1	2.5
Difference in D max			X	---	Avg. of 2	---	.01	S-1	2.5

1/ If failure is indicated, report either description of failure or numerical point of failure
2/ Test reports shall include all values on which results are based
3/ Not less than 7 sample units shall be tested

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4.4 Inspection of preparation for delivery. Sample items and packs shall be selected and inspected in accordance with PPP-P-16.

5. PREPARATION FOR DELIVERY

*5.1 Preservation and packaging. The preparation of the developer for delivery, in 1 and 5 gallon sizes, or other size as required in 3.4, shall be in accordance with PPP-P-16, except as specified (see 6.2). Commercial puncture type cans are also acceptable. Packaging shall be level A, B, or C or PPP-P-16 as specified in the contract or order (see 6.2).

*5.1.2 Unit packaging. Unless otherwise specified by the procuring activity (see 6.2), unit packaging shall be in accordance with PPP-P-16, Level A, Type II, Class 1, or a commercial puncture type can. The container shall be constructed of a material that is inert to the chemicals used in the developer.

*5.1.3 Unit container marking. Unless otherwise specified by the procuring activity, each unit container shall be durably and legibly labeled by Lithographic or other suitable means. If gummed labels are used, they shall be affixed with water resistance adhesive. The adhesive shall not smear or blur the markings.

*5.1.3.1 The following shall be on the label:

a. Item identification
b. Directions for mixing and use
c. Manufacturer's name and code number
d. Net weight

*5.1.3.2 The following shall be either lithographed directly on the label or clearly printed on the lid:

a. Specification number
b. Type number
c. Stock number
d. Contract number
e. Date of manufacture
f. Expiration date

*5.1.3.3 In accordance with International Agreements (see 6.3), the final mixed volumes shall be indicated in U.S. gallons or ounces, imperial gallons or ounces, and litres or cubic centimeters. The expiration date shall be stated as: "To be used before ..." Special storage conditions, if required, shall also be indicated.

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*5.1.3.4 Date of manufacture. The date of manufacture shall be the month and year of manufacture of the material specified herein. Example: "(Feb 66)" would refer to February 1966.

5.2 Packing. Packing shall be level A, B, or C of PPP-P-16, as specified in the contract or order (see 6.2).

5.3 Marking. Marking shall be in accordance with PPP-P-16. In addition, the following precautionary marking shall be included on the exterior container.

PHOTOGRAPHIC CHEMICALS

STORE IN A DRY COOL PLACE AWAY FROM
RADIOACTIVE MATERIALS

6. NOTES

*6.1 Intended use. The developer covered by this specification is intended for use in processing film where a requirement exists for either high contrast, medium contrast or rapid process high contrast.

*6.2 Ordering data. Procurement documents should specify:

a. Title, number and date of this specification.
b. Nominal size (1-,5-, or 20-gallon or size specified by the procuring activity.
c. Quantity required
d. Type of developer required (see 1.2.1).
e. Applicable levels of preservation, packaging, and packing required (see 5.1, 5.2 and 5.4).
f. When allowable tolerance of total net weight of the individual chemicals shall vary from ± 3 percent (see 3.2).

*6.3 International standardization. Certain provisions of this specification (3.2) (5.3) are the subject of International Standardization Agreements (ASCC Air Std 52/3, ASCC Air Std 52/5A, STANAG 3182, STANAG 3187). When amendment, revision or cancellation of this specification is proposed, the departmental custodians will inform their respective departmental standardization offices so that appropriate action may be taken respecting the international agreement concerned.

*6.4 The margin of this specification is marked with an asterisk to indicate where changes (additions, modifications, corrections, deletions) from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and

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Custodians:
Army - EL
Navy - AS
Air Force - 70

Preparing activity:
Air Force - 70

Project No 6750-0312

Reviewer activities:
DGSC - GS
Navy - MC
Army - EL

User activities:
Army - MI, AV
Navy - YD

International interest (see 6.3)

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

OMB Approval
No. 22-R255

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☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE DOCUMENT CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

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☐ YES ☐ NO *(If "Yes", in what way?)*

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SUBMITTED BY *(Printed or typed name and address - Optional)*

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